# A Drift-Adaptive Hybrid Online Streaming Intrusion Detection Scenario for Electric Vehicle Charging Systems

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**Abstract.** Electric Vehicle Charging Systems are being included as an important part of Smart Grid Systems as well as IoT-based systems. It is quite obvious that the threat level for complex cyber-physical threats is also increasing. It is seen that conventional security, as well as forensic analysis systems, are more of reactive systems, which are not efficient for real-time security analysis. In this paper, we propose a real-time forensic-ready anomaly-based network security system for EVCS networks using real-time machine learning and blockchain technology. In this paper, a hybrid model of Hoeffding Tree and Adaptive Random Forest is proposed for efficient learning and classification. To detect temporal patterns of attacks, an Long Short-Term Memory (LSTM) based component for analyzing the temporal patterns of the attacks, which captures the sequential dependencies of the EVCS network. Once the anomalies are identified, the proposed approach offers in-depth forensic logging, hashing, and secure storage of the evidence in an efficient blockchain framework for guaranteed preservation of tamper-proof evidence. To validate the efficacy of the proposed approach, the proposed approach is tested using the CICEVSE2024 dataset, simulating realistic EVCS network scenarios. From the experimental results, it is clear that the proposed approach offers high accuracy, along with other performance matrices measure values, validating the efficacy of the proposed approach. This approach, through the integration of real-time detection capabilities, offers an efficient security-enhanced approach for the next generation of EVCS.

**Keywords:** Electric Vehicle Charging Systems (EVCS) · Cybersecurity · Network Security · Real-Time Anomaly Detection · Hoeffding Tree · Adaptive Random Forest · Long Short-Term Memory · Streaming Machine Learning · Concept Drift · Blockchain · Digital Forensics · Smart Grid Security.

## 1 Intorduction

The Increasing adoption of Electric Vehicle Charging Systems (EVCS) in smart grids and IoT-based configurations is revolutionizing the current energy sector.

The charging systems assist in the efficient transfer of energy and address the increasing demand for electric vehicles. However, as EVCS link charging points, control servers, grid management systems, and user interfaces, they also introduce a wide range of cyber attacks. Denial-of-service attacks, data manipulation, false data injection, and unauthorized access can bring down the system, damage data, and disrupt the energy grid. The critical role EVCS play in today's energy infrastructure, protecting these systems and ensuring they can recover from disruptions is now the highest priority in research on cyber and network security. The traditional approach to security in EVCS scenarios is generally reactive, with a focus on forensic analysis after an attack. These approaches are useful for understanding what happened after the attack but are ineffective in preventing and mitigating threats in rapidly changing , real-time environments. Futhermore, many traitional machine learning algorithms are batch-based, with assumptions about fixed data patterns. However, in real-world EVCS networks, traffic patterns and attack dynamics evolve over time, resulting concept drift. Traditional detection systems are not equipped to handle such dynamic environments, making the need for adaptive and incremental learning in real-time data analysis pressing.

Blockchain technology provides a decentralized and cryptographically secure method for mantaining logs in a trustworthy manner. By creating cryptographic hashes and chaining records together, blockchain technology makes it difficult to manipulate stored data without being detected. Integrating blockchain technology for log data preservation with adaptive anamaly detection opens the door for a forensic-ready security solution. This solution not only provides rapid identification of malicious activity but also provides tamper-proof storage of forensic data, which increases trust and accountability within EVCS environments. This paper introduces a real-time forensic-ready anomaly detection solution for EVCS networks. This solution integrates a streaming ensemble model, including Hoeffding Tree and Adaptive Random Forest, to provide low-latency detection and effective concept drift handling. An optional LSTM component can be integrated to analyze temporal patterns and sequential attack patterns in EVCS traffic. Once anomalies are identified, the system provides in-depth forensic logging, hashes the logs using SHA-256, and stores them in a simplified blockchain format to ensure immutability and integrity. The efficacy of the proposed solution will be evaluated based on its application to the CICEVSE2024 dataset, which represents a practical case of traffic and attack patterns in the EVCS network. The proposed solution will be found viable for integrating streaming machine learning and blockchain-based secure logging.

## 2 literature survey

Al-E'mari et al. [1], The Electric Vehicle Charging Systems (EVCS) is more prone to cybersecurity attacks as it is being integrated with smart grids and IOT networks. This is one of the major issues concerning the cybersecurity of

the Electric Vehicle Charging Systems. Research has been conducted regarding anomaly detection at the network level, while it has not been explored at the host level. In this paper, a new forensic analysis framework is proposed, which utilizes host-level information such as system logs, kernal messages, and Hardware Performance Counters (HPC), to identify and analyze complex cyberattacks such as cryptojacking, Denial-of-Service (DoS), and reconnaissance attacks on EVCS. By conducting compprehensive forensic analysis and machine learning techniques, the proposed framework significantly outperforms the existing literature, achieving an accuracy of 98.81. The results provide new insights into the unique behavioral characteristics of various specific cyber threats, which can help develop better cybersecurity policies and recommendations for effective EVCS infrastructure protection.

Shin et al. [2] addresses as Electric Vehicles (EVs) gain speed towards mass adoption, the charging infrastructure is also expanding and becoming more complex. As growth is accelerating, cybersecurity challenges are becoming more critical. A big chunk of the solution domain still needs to be explored. There is an urgent need to develop a structured data-driven approach to digital forensics to deal with these issues. It is difficult to identify and attribute the above-mentioned scenarios, which has given birth to a new data-driven digital forensics problem-solving approach. It is proposed to develop this new approach using open-source intelligence and state-of-the-art modeling techniques in a cyber-physical system architecture consisting of three layers to identify digital evidence. This concept is linked to a variety of cybersecurity issues and is expected to enhance digital forensics capabilities, comply with evolving cybersecurity regulations, and ensure the security and trustworthiness of digital forensic practices.

Alcarza et al. [3] argues that Industry 4.0 is transforming the energy industry by increasing automation, production, and distribution. This trend has also accelerated the development of grid-connected charging infrastructure. These charging stations are capable of following standardized frameworks such as the Open Charge Point Protocol (OCPP). The most recent version, OCPP v2.0.1, has incorporated improved security measures at both the device and communication levels to mitigate the weakness identified OCPP v2.0.1 to determine whether its new features could still be vulnerable to certain cyber and physical attacks, particularly when charging infrastructure is interconnected with microgrids. To achieve this, we shall first modify the STRIDE threat analysis framework, which will be tailored to assess threats based on control and energy, and then combine it with DREAD. Our results indicate that despite the improvements in OCPP v2.0.1, security threats still exist, which require improved security measures in the future.

Mastoi et al. [4] argues that the world is undergoing a paradigm shift: the transportation from fossil fuels to zero or ultra-low emission vehicles. The adoption of electric vehicles requries the development of a charging infrastructure

network, consisting of charging stations that utilize information technology, effectively managed energy resources, and government strategies that promote development. This article presents the key considerations in the planning of electric vehicle charging infrastructure. It provides information on planning strategies and technological developments that can enhance the design and implementation of charging stations. It presents a general perspective on the current state of the electric vehicle industry, the impact of EVs on the power grid, and the most effective development of EV charging infrastructure. In particular, it examines research on the development of charging station infrastructure, the difficulties associated with the development process. In the future, the article examines future trends, including the use of renewable energy sources and the role of EVs in the development of the power grid technology.

## 3 Proposed Methodology

The proposed framework aims at providing a real-time, forensic- ready anomaly detection system for Electric Vehicle Charging Systems (EVCS) that function in a dynamic smart grid environment. The proposed framework combines various aspects, including real-time classification using a hybrid incremental stream learning approach, temporal anomaly detection using an LSTM-based autoencoder, adaptive concept drift detection, and tamper-proof forensic logging using a hash-chain evidence preservation mechanism. The proposed methodology is specifically designed to handle high-velocity EVCS network traffic while ensuring scalability, adaptability, and forensic integrity.

### 3.1 Dataset Description

The aim of the EV Charger Attack Dataset 2024 (CICEVSE2024) seeks to move the limit of cybersecurity research in the area of electric vehicle charging stations. To achieve this, it provides a comprehensive dataset that enclose not only normal and safe functionality bu talso different attack types. These types include both network and host attacks on the EVSE, whether it is idle or in the middle of a charging process. To achieve this, an experimental testbed was developed. This testbed is comprised of a Level 2 functional EV charging (EVSE-A) and a number of Raspberry Pi Systems, as well as the communication equipment. This testbed is a replica of the key components of an actual electric vehicle charging station and provides the capability to collect host-level data on all the systems involved.

The dataset set consists of processed and labeled CSV files, which are generated from the original samples. The dataset tracks host-level events, including HPC and Kernel events, over larger time intervals and distinguishes between attack and benign samples. The Dataset describes the various scenarios and corresponding attack types from the CICEVSE2024 dataset, as well as their impact on the CIA triad: Confidentiality, Integrity and Availability. The dataset contains

both benign and attack samples, with host-level events recorded over large time intervals. The scenarios are identified as either normal functioning(benign) or a particular type of cyber attack. The different types of attacks include cryptojacking, vulnerability scanning, and different types of DoS attacks. The dataset is useful in examining the behavior of the system and identifying malicious activity, which is important in the development of effective cybersecurity research and development.

### 3.2 Preprocessing

The dataset CICEVSE2024 was subjected to a preprocessing stage to ensure it was ready for live-streaming classification and time-based anomaly detection for EV charging stations.

**Data Cleaning** In this stage, irrelevant features, redundant identifiers, and uninformative data were removed to reduce dimensionality.

**Feature-Label Separation** The dataset was divided into feature matrices X and a target label vector y, with the labels representing normal or attack states.

**Label Encoding** Label encoding was done to ensure that all labels were ready for use in machine learning algorithms, as were all the categorical features.
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**Categorical Feature Encoding** The following categorical input features are encoded into numerical representations to maintain structural consistency among the models.

**Missing Value Handling** The Missing values, null values, or undefined values are handled using imputation techniques or row filtering to prevent bias and instability during the training.

**Feature Scaling**

- Standard Scaling is applied for both the streaming classifiers (Adaptive Random Forest and Hoeffding Tree) to stabilize incremental learning.

– Min–Max Normalization is applied for the Long Short-Term Memory component to ensure stable gradient convergence and balanced feature ranges.

**Streaming Data Formatting** The preprocessed dataset is converted into an instance-wise streaming format compatible with incremental learning using the River tool framework for enabling real-time processing.

**Temporal Window Construction** The fixed-length sequential windows (time step t=5) are generated to create three-dimensional tensors for Long Short-Term Memory processing for enabling the detection of sequential attack patterns.

**Chronological Ordering** The following dataset is maintained in the sequential order without random shuffling to preserve temporal integrity and simulate real-world Electric Vehicle Charging Station network traffic conditions.

**Final Validation** The following processed dataset s verified for consistency, numerical stability, and compatibility with both adaptive ensemble classifiers and deep sequential models used in the proposed forensic-ready anomaly detection framework.

This data preprocessing workflow ensures data integrity, stability, and consistency. It integrates smoothly with the streaming and deep learning components of the proposed EVCS security framework.

### 3.3 Hoeffding Tree (HT)

In the proposed architecture, the Hoeffding Tree (HT) is used as a fast decision tree classifier for real-time data analysis of EVCS network traffic. As EVCS environments are known to produce a constant stream of data, HT processes one instance at a time and adjusts its model parameters on the fly rather than having to rebuild its model from scratch. HT relies on the Hoeffding bound to select statistically significant splits based on the features, which essentially means rapid, memory-efficient learning with virtually zero latency, making it ideal for the high-speed charging network environment. Additionally, HT's ability to learn incrementally makes it an excellent candidate for adaptive EVCS networks that experience changes in traffic patterns over time.

### 3.4 Adaptive Random Forest (ARF)

Adaptive Random Forest (ARF) is the primary ensemble method for streaming classification to improve the robustness of detection and handle concept drift in the dynamic EVCS environment. ARF consists of multiple incremental decision trees that are trained using online bagging, where each tree is trained on different portions of the streaming data. It has drift detection methods to replace the trees when their performance degrades with the evolution of attack patterns. In this context, ARF improves the stability of classification, reduces the number of false positives, and adapts to the changing cyber-physical attack patterns in the EVCS networks. When HT is combined with ARF using soft voting, the approach achieves better detection accuracy and robustness compared to single-model solutions.

### 3.5 Long Short-Term Memory (LSTM)

The Long Short-Term Memory (LSTM) module is incorporated to better understand the temporal nature of attacks, which involve sequences of actions that a single-shot classifier cannot identify. EVCS attacks are typically multi-step and time-variant, making it essential to model the temporal component. In our system, an autoencoder with an LSTM module is used to analyze fixed-size sequences of traffic, attempt to reconstruct them, and compute the reconstruction error. If the error exceeds a certain threshold, then the sequence is considered anomalous. This method is useful for detecting slow, low-rate attacks that would otherwise be overlooked by the classifier. To conclude, the addition of the LSTM component enhances the performance of the system.

### 3.6 Adaptive Windowing Algorithm (ADWIN)

It continuously monitors for a concept drift to ensure that the system remains reliable. In real life, it is expected that traffic and types of attacks change over time for a variety of reasons. Therefore, it would not be effective to have a static system. ADWIN monitors the errors that both the random forest algorithm and the Hoeffding tree algorithm produce in classifying instances. Both algorithms use a sliding window of instances that is constantly changing as new instances are added. If new instances differ from those in the window by more than a certain amount, it indicates a drift, which would require a system update to ensure that it remains effective in a smart grid system.

### 3.7 Hash-Chained Logging for Forensic Integrity

In order to improve forensic readiness and ensure that data is tamper-proof, it uses a hash chain for logging data, reminiscent of blockchain technology. In case

of any deviation, it will log all details of that event, including when it occurred, which attributes were involved, what the prediction was, and what the anomaly score was. It will then hash that data with SHA-256 and create a chain. If any of that data is modified, it will shift the entire chain because it is all linked. It will make it difficult to tamper with the log data and will turn it into a forensic-ready security solution for the next generation of EVCS.

## 4 Experimental Simulation

The proposed framework for real-time, forensic-ready anomaly detection was tested by the team through an experimental simulation, where they tested how well the framework identifies cyber threats in Electric Vehicle Charging Systems (EVCS). To implement the framework, the team used the Python programming language. To develop the streaming machine learning models, the River framework was used, while the LSTM-based autoencoder model was developed using TensorFlow/Keras. To evaluate the proposed framework, the team used the CICEVSE2024 dataset, which simulates realistic network traffic for EVCS, including both normal behavior and different types of attacks. To evaluate the framework in an environment that simulates real-world EVCS, the team used the sequential processing approach, as opposed to batch learning. Each piece of data was processed individually using the hybrid classifier, which is based on the combination of Adaptive Random Forest (ARF) and Hoeffding Tree (HT). To evaluate the framework, the team used the test-then-train approach, where predictions are made before updating the model, as shown in Fig. 1. To detect concept drift, the team used ADWIN, where prediction errors are constantly monitored, as shown in Fig. 1.

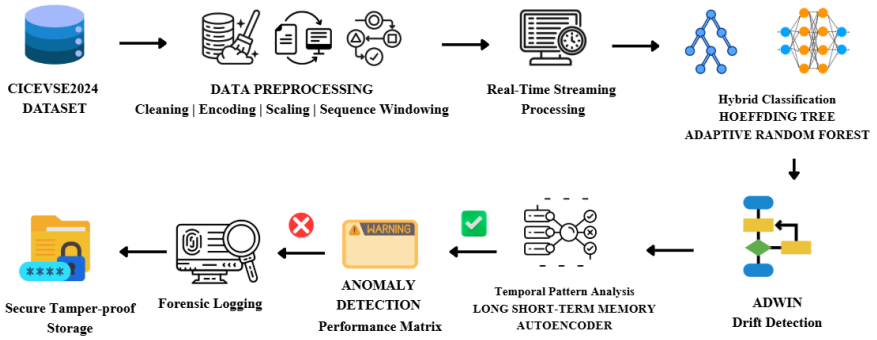


**Fig. 1.** System Architecture

To deal with timing issues, we divided our data set into pieces that were fed to our LSTM autoencoder model. Our idea was quite simple: if our model's prediction was below a certain threshold, it was considered an anomaly. This approach helped us detect time-based attacks that other models would otherwise miss. For log integrity, we made sure that each log entry was hashed using SHA-256 encryption before it was linked to the previous log entry. We also utilized standard performance metrics for our model: Accuracy, Precision, Recall, F1 Score, as well as accuracy improvement over time.The results confirmed high detection performance, effective drift adaptation, reliable temporal attack identification, and secure forensic logging, demonstrating the framework's suitability for next-generation EVCS security applications as shown in the Fig.1.

## 5 Results and Discussion

The proposed real-time framework was tested using the CICEVSE2024 data set in a streaming learning setup. The Adaptive Random Forest (ARF) with a Hoeffding Tree (HT) ensemble showed very stable and accurate classification results throughout the data stream. The streaming accuracy results showed some minor variations during the initial learning phase, which is normal for incremental learning models because of cold-start learning. However, as the number of samples increased, the accuracy values stabilized quickly and remained at a constant level of around 99.95% towards the end of the data stream as shown in the Fig.2. This is an indication of strong online learning abilities and successful management of dynamic data patterns. The inclusion of the ADWIN drift detector in the ARF algorithm helped to make the system resistant to possible concept drifts, thus ensuring stable performance throughout the entire dataset. A total of 63,869 anomalies were identified in the streaming process, which shows that the system is very sensitive to unusual behavioral patterns in EVCS network traffic as shown in the Fig.2.

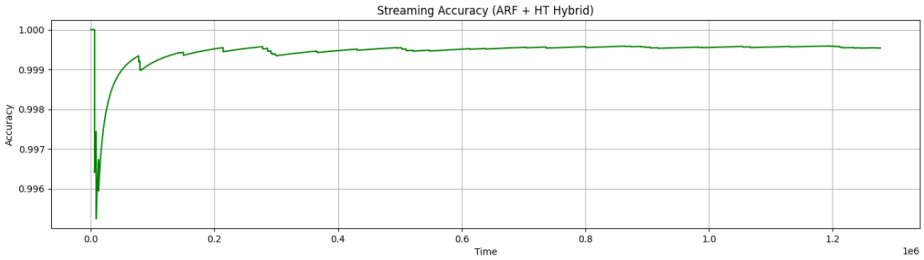


**Fig. 2.** Streaming Accuracy of Adaptive Random Forest (ARF) and Hoeffding Tree (HT)

The LSTM autoencoder further improved the model by studying the temporal behavior of the data via the reconstruction error (MSE) as shown in the Fig.3. The training loss steadily reduced with each epoch, ensuring proper learning of the normal traffic behavior. The anomaly threshold, calculated by the 95th percentile of the reconstruction error, properly separated normal data from anomalous data. The graph of the reconstruction error clearly indicates the points beyond the threshold, representing anomalous traffic patterns. These points were then recorded using a hash chain method to provide tamper-proof forensic storage. The integration of adaptive streaming classification and temporal anomaly detection proves the efficacy of the proposed integrated model. The results clearly demonstrate that the system has high detection accuracy, strong adaptability, and sound forensic integrity, making it appropriate for securing next-generation EVCS networks.

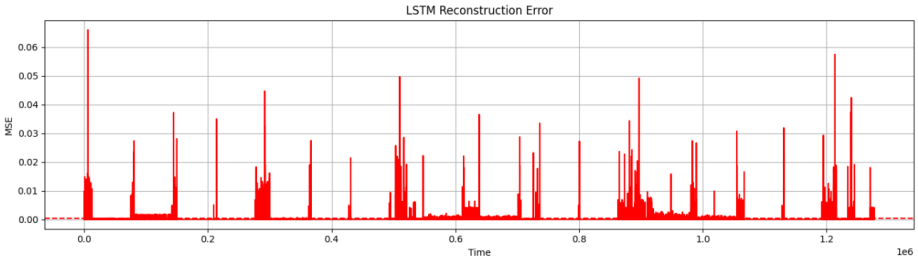


**Fig. 3.** Long Short-Term Memory Reconstruction Error

## 6 Conclusion

This research proposed an "Autonomous Real-Time Forensic Security Framework for Electric Vehicle Charging Systems (EVCS)" to overcome the shortcomings of the conventional post-mortem and network-centric forensic analysis. The proposed framework combines Adaptive Random Forest (ARF) with Hoeffding Tree (HT) classification for real-time streaming data classification and an LSTM autoencoder for temporal anomaly detection. The proposed framework combines incremental machine learning with deep temporal analysis to facilitate continuous monitoring, early attack warning, and adaptive learning in dynamic EVCS settings.

The experimental outcome has shown that the hybrid ARF-HT model maintained a stable streaming classification accuracy of 99.95%, and the LSTM autoencoder successfully identified temporal anomalies based on the reconstruction error threshold. The proposed framework has also successfully identified a substantial number of anomalies and maintained the stability of the classification

performance throughout the data stream. Moreover, the integration of tamper-proof logging further improves the forensic validity of the proposed framework.

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